

TacMagPie: A Fast, High-Fidelity Soft Magnetic Tactile Sensor Simulator for Sim-to-Real Robotic Manipulation

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Abstract—Tactile perception is a cornerstone of large-scale robot learning. Among various tactile modalities, soft magnetic tactile sensors (sMagTac) stand out by offering rich 3D contact force measurements with exceptional durability and sensitivity. However, obtaining large-scale physical data from sMagTac remains prohibitively costly. While simulation offers a natural remedy, it is severely hindered by the complex magneto-elastic coupling that governs force transduction in these sensors. Existing methods are limited to slow finite element method (FEM) or simplified analytical models based on restrictive assumptions, neither of which supports real-time interaction with modern robot simulators. We present *TacMagPie*, a real-time simulator that achieves fast, physically consistent simulation of sMagTac. *TacMagPie* combines Moving Least Squares Material Point Method (MLS-MPM) for hyperelastic modeling of silicone elastomers with Differential Magnetic Microparticle Method (DMM) for accurate magnetic flux computation. Quantitative evaluation shows strong agreement with real-world measurements, achieving a PSNR of 26.21 dB and an SSIM of 0.68, indicating strong structural agreement in contact-position-indexed Hall-response maps. At the same time, *TacMagPie* runs in under 1 ms per simulation step, 3–4 orders of magnitude faster than FEM, making it suitable for large-scale synthetic tactile data generation. A downstream contact-localization study further demonstrates its potential as a practical simulation backend for learning-based magnetic tactile perception.

I. INTRODUCTION

Tactile perception is the cornerstone of dexterous manipulation in both biological systems and embodied artificial intelligence. While vision enables agents to perceive the geometry and semantics of their surroundings, touch provides direct access to contact physics such as texture, hardness, and the onset of slipping, which are indispensable for fine-grained action control [1]. In the era of large-scale robot

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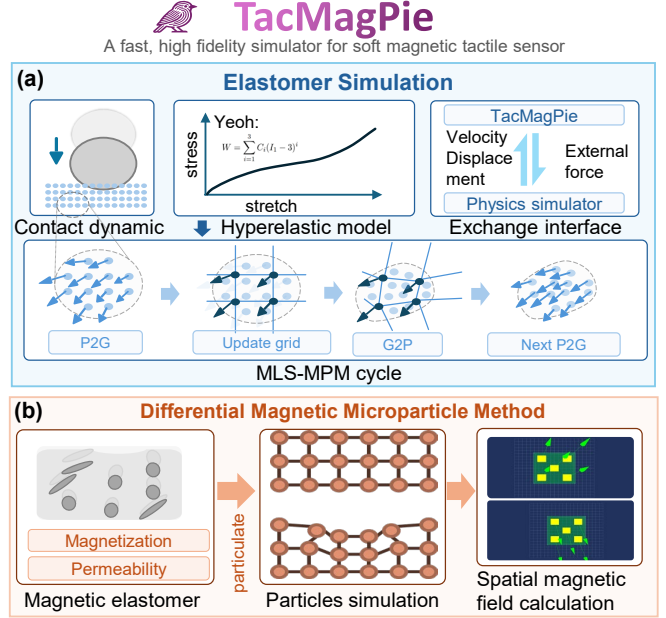


Fig. 1. *TacMagPie* comprises two core components: (a) MLS-MPM-based elastomer deformation simulator for efficient hyperelastic modeling, and (b) DMM for accurate magnetic flux computation.

learning, particularly with the advent of Vision-Language-Action (VLA) models, tactile feedback is increasingly recognized as essential for resolving contact dynamics that are inherently ambiguous or unobservable from vision alone [2]. This growing recognition has driven the demand for scalable tactile data and reliable tactile simulation tools.

TABLE I
REPRESENTATIVE COMPARISON OF TACTILE SENSING MODALITIES FOR ROBOTIC MANIPULATION

Modality	3D Force	Spatial Resolution	Compactness & Robustness	Sim-to-Real Readiness
Piezoresistive [3]	✗	Medium	Low	Low
Capacitive [4]	✗	Medium	Medium	Low
Optical [5]	✓	High	Low	High
Magnetic [6]	✓	Med-High	High	<i>Underexplored</i>

* This table summarizes representative trends rather than exhaustive modality-level properties, as practical performance can vary with specific sensor designs and implementation details.

Various tactile transduction principles have been explored to digitize physical interactions, as summarized in Table I. While existing solutions suffer from inherent limitations, soft magnetic tactile sensors (sMagTac, shown in Fig. 2(a))

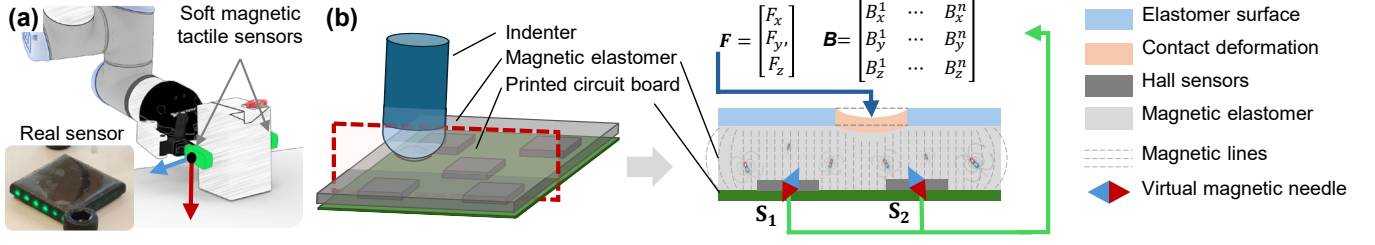


Fig. 2. Soft magnetic tactile sensors. (a) Illustration of tactile sensing in robotic manipulation. (b) Working principle of sMagTac. The indenter causes elastic deformation, and a Hall sensor array on the measurement circuit acquires a 3D magnetic field matrix, enabling direct access to decoupled pressure and shear forces.

represent a compelling balance of sensing performance, durability, and compactness for robotic manipulation. As shown in Fig. 2(b), sMagTac exploit magnetic field variations induced by elastomer deformation to measure three-dimensional contact forces. They offer high sensitivity to shear forces, excellent durability, and a compact form factor. Moreover, due to the separation between the soft elastomer and the measurement circuitry, these sensors exhibit strong wear resistance in real-world applications.

However, despite their hardware superiority and increasing adoption, the corresponding simulation research remains significantly underdeveloped. Unlike optical sensors that leverage established ray-tracing pipelines [7], [8], magnetic sensors rely on complex magneto-elastic coupling, where the displacement of embedded magnetic microparticles is intricately tied to the surrounding elastic deformation [9]. As a result, simulation has long relied primarily on the finite element method (FEM) and similar approaches. FEM is known to be physically rigorous but computationally prohibitive for real-time use: a single simulation step can take seconds to minutes [10]. Analytical approximations achieve fast inference but are tailored to specific sensor geometries, failing to generalize across designs or capture dense displacement fields [11]. Therefore, there remains a critical need for a simulation framework that is both computationally efficient and physically consistent, while remaining adaptable to different sensor geometries and contact boundary conditions.

To bridge this gap, we propose *TacMagPie*, a real-time simulation framework for reconstructing magnetic tactile responses in sMagTac. As shown in Fig. 1, the simulator models elastomer deformation using MLS-MPM and resolves magneto-elastic coupling via a DMM formulation. By reformulating the coupled field computation, *TacMagPie* reduces the runtime of a single simulation step from seconds to under 1 ms with GPU acceleration. Despite its efficiency, the simulator remains physically consistent: the synthesized sensing responses closely match real-world measurements, achieving a PSNR of 26.21 dB and an SSIM of 0.68, indicating high fidelity comparable to graphics-level similarity standards.

The main contributions are summarized as follows:

- Proposal of the DMM method, an effective approach that computes magnetic flux density from discretized magnetic microparticles, enabling real-time resolution

of magneto-elastic coupling;

- Development of *TacMagPie*, a physically consistent real-time simulation framework for sMagTac;
- Validation of its physical consistency and computational efficiency against real-world measurements, demonstrating its potential for synthetic tactile data generation and learning-based magnetic tactile perception.

II. RELATED WORKS

A. Elastomer Simulation

Soft magnetic tactile sensors transduce contact forces through the deformation of the elastomer, meaning the fidelity of simulation is fundamentally determined by how accurately the elastomer is modeled. Most mainstream robotics simulators, such as PyBullet [12], MuJoCo [13] and Isaac Gym [14], are designed primarily for rigid-body dynamics and offer only simplified mass-spring models, which are insufficient for the large, nonlinear deformations characteristic of soft silicone.

FEM can accurately simulate hyperelastic materials, but it requires fine meshes, suffers from mesh tangling under large deformations, and is computationally prohibitive for real-time or large-scale data generation [15]. Meshfree alternatives such as SPH [16], [17] and the original MPM [18] avoid remeshing by representing the material as Lagrangian particles, yet classical MPM remains prone to cell-crossing instabilities under high strain. MLS-MPM [19] resolves these issues through a moving least squares velocity reconstruction that recovers APIC momentum transfer [20] and consistent stress divergence in a unified formulation. This yields unconditional stability under large deformations, natural GPU parallelism, and dense per-particle displacement tracking, which provide exactly the kinematic information needed for downstream magnetic field inference. We therefore adopt MLS-MPM as the mechanical backbone of our framework.

B. Simulation of Vision-Based and Magnetic Tactile Sensors

For vision-based tactile sensors, simulation is largely mature: tools like TACTO [7] and Taxim [8] reformulate the problem as a rendering task, achieving high visual fidelity via graphics pipelines [21]. Magnetic tactile sensors, however, require solving magneto-elastic coupling, which falls outside the capabilities of mainstream simulation engines.

Existing approaches occupy two extremes. FEM-based tools such as COMSOL [22] faithfully capture magneto-elastic coupling but require seconds per step, precluding use in learning loops. Analytical approximations [11] and lookup tables [23] achieve real-time rates but assume simplified geometries, failing to reproduce distributed displacement fields under shear or torsion. To bridge this gap, we propose the Differential Magnetic Microparticle method (DMM), which couples the MLS-MPM deformation field with an analytical dipole solver. Each embedded magnetic particle is continuously tracked by the MPM simulation, and the resulting sensor signal is computed at every step, retaining physical fidelity while remaining efficient enough for large-scale robot learning.

III. METHODS

Our simulation pipeline consists of two decoupled stages: (1) Sec. III-A introduces how *TacMagPie* simulates elastomer deformation; (2) Sec. III-B proposes a method for magnetic flux computation. Finally, Sec. III-C describes how *TacMagPie* integrates with a physics simulator as a plug-and-play backend.

A. Elastomer Deformation Physics

TacMagPie employs the Moving Least Squares Material Point Method (MLS-MPM) [19] to simulate silicone elastomer deformation (Shown as Fig. 1(a)). MPM represents the material as a collection of Lagrangian particles while using a background Eulerian grid solely as a computational scratch space for solving momentum equations. The MLS-MPM loop consists of particle-to-grid (P2G) transfer, grid update, and grid-to-particle (G2P) transfer. Unlike FEM, where large deformations can cause mesh tangling and negative element volumes that degrade stress accuracy, MPM particles never distort—they simply advect through the fixed grid, making the method naturally robust to the extreme deformations that occur during tactile contact.

Silicone is hyperelastic: its stress response is nonlinear and depends only on the current deformation state. We adopt the Yeoh model [24], which expresses strain energy density as a polynomial in the first invariant $I_1 = \text{tr}(\mathbf{F}^\top \mathbf{F})$:

$$W = \sum_{i=1}^3 C_i (I_1 - 3)^i, \quad (1)$$

where $\mathbf{F}$ is the deformation gradient and C_1, C_2, C_3 are material constants fitted to the target silicone. The higher-order terms capture the strain-stiffening behavior under deep indentation that simpler Neo-Hookean models miss, while remaining more computationally lightweight than Ogden or Mooney-Rivlin formulations. The first Piola-Kirchhoff stress follows as $\mathbf{P} = \partial W / \partial \mathbf{F}$.

At each time step, particle quantities are transferred to the background grid. Specifically, MLS-MPM [19] augments each particle with an affine momentum matrix $\mathbf{C}_p$, enabling

a more accurate and stable transfer:

$$(m\mathbf{v})_i = \sum_p w_{ip} \left[m_p (\mathbf{v}_p + \mathbf{C}_p (\mathbf{x}_i - \mathbf{x}_p)) - \Delta t V_p \mathbf{P}_p \mathbf{F}_p^\top \nabla w_{ip} \right], \quad (2)$$

where w_{ip} is the interpolation weight between particle p and grid node i , and the stress term applies internal elastic forces directly during transfer without a separate assembly step.

After applying boundary conditions on the grid, where bottom-layer particles are fixed as Dirichlet constraints to replicate the sensor's physical mounting, updated velocities are interpolated back to particles:

$$\mathbf{v}_p = \sum_i w_{ip} \mathbf{v}_i \quad (3)$$

$$\mathbf{C}_p = \frac{4}{\Delta x^2} \sum_i w_{ip} \mathbf{v}_i (\mathbf{x}_i - \mathbf{x}_p)^\top. \quad (4)$$

To model the viscoelastic energy dissipation of silicone and ensure convergence to equilibrium after contact, a damping factor $\alpha \in (0, 1)$ is applied to all grid velocities prior to G2P transfer: $\mathbf{v}_i \leftarrow \alpha \mathbf{v}_i$. The deformation gradient is then updated multiplicatively to preserve large-deformation history:

$$\mathbf{F}_p^{n+1} = (\mathbf{I} + \Delta t \mathbf{C}_p) \mathbf{F}_p^n, \quad (5)$$

and particles are advected as $\mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \Delta t \mathbf{v}_p$. The resulting per-particle displacement field is passed directly to the magnetic field solver described in the next section.

B. Magnetic Field Computation

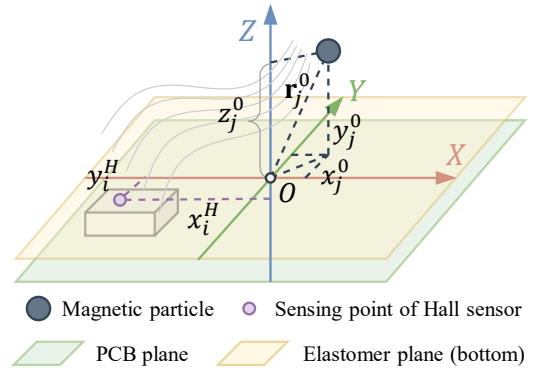


Fig. 3. Magnetic field model of a uniformly magnetized elastomer. A differential volume element at position $\mathbf{r}_j^0$ contributes a differential magnetic flux density $d\mathbf{B}_j$ at the sensor location $\mathbf{r}_{s,k}$.

We propose an effective Differential Magnetic Microparticle (DMM) method that efficiently reconstructs magnetic flux variations from elastomer deformation. The elastomer is doped with magnetic particles, modeled as a discrete set of N_m magnetic dipoles. Let the j -th microparticle possess a magnetic moment $\mathbf{m}_j \in \mathbb{R}^3$. The magnetic flux density $\mathbf{B}$ generated by this dipole at a sensor position $\mathbf{r}_s$ is given by the analytical solution to the magnetostatic equations:

$$\mathbf{B}(\mathbf{r}_s, \mathbf{r}_j, \mathbf{m}_j) = \frac{\mu_0}{4\pi} \left(\frac{3(\mathbf{m}_j \cdot \mathbf{r}_{js})\mathbf{r}_{js}}{\|\mathbf{r}_{js}\|^5} - \frac{\mathbf{m}_j}{\|\mathbf{r}_{js}\|^3} \right) \quad (6)$$

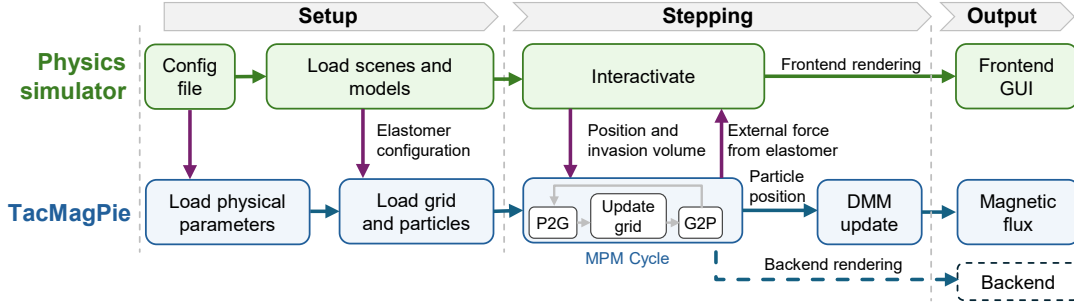


Fig. 4. Interaction pipeline between *TacMagPie* and a physics simulator (e.g., MuJoCo). Both load configurations from a shared config file, initializing the simulation environment and sensor parameters (e.g., hyperelastic coefficients, Hall sensor positions). At each timestep, *TacMagPie* receives boundary conditions from the simulator, advances particle and grid states via the MPM loop, and computes the magnetic field at each Hall sensor using DMM method. Reaction forces are simultaneously fed back to the simulator. *TacMagPie* outputs magnetic field observations synchronized with the simulator.

where μ_0 is the vacuum permeability, and $\mathbf{r}_{js} = \mathbf{r}_s - \mathbf{r}_j$ is the displacement vector from the particle to the sensor.

The core innovation of DMM is tracking the differential evolution of the field during deformation. In the initial state, the particle is at $\mathbf{r}_j^0$. Upon deformation, its position updates to $\mathbf{r}_j = \mathbf{r}_j^0 + \mathbf{u}(\mathbf{r}_j^0)$, where $\mathbf{u}(\mathbf{r}_j^0)$ is obtained via the particle displacement in Sec. III-A. The total magnetic field measured by the k -th Hall sensor $\mathbf{S}_k$ is the superposition of contributions from all microparticles:

$$\mathbf{B}_k^{\text{total}} = \mathbf{B}_{\text{earth}} + \sum_{j=1}^{N_m} \mathbf{B}(\mathbf{r}_{s,k}, \mathbf{r}_j, \mathbf{m}_j) \quad (7)$$

Directly recomputing Eq. (7) at every step is computationally expensive. Instead, we compute the incremental field variation $\Delta \mathbf{B}_k$:

$$\Delta \mathbf{B}_k = \sum_{j=1}^{N_m} [\mathbf{B}(\mathbf{r}_{s,k}, \mathbf{r}_j^0 + \mathbf{u}(\mathbf{r}_j^0), \mathbf{m}_j) - \mathbf{B}(\mathbf{r}_{s,k}, \mathbf{r}_j^0, \mathbf{m}_j)] \quad (8)$$

This formulation improves numerical stability and allows for efficient parallelization. Finally, we optionally add zero-mean Gaussian noise to mimic short-term electronic measurement noise, while leaving complex effects such as hysteresis and temperature drift for future extensions:

$$\hat{\mathbf{B}}_k = \mathbf{B}_k^{\text{ref}} + \Delta \mathbf{B}_k + \mathcal{N}(\mathbf{0}, \Sigma_{\text{noise}}) \quad (9)$$

where $\mathbf{B}_k^{\text{ref}}$ is the calibrated baseline reading.

C. Modular Architecture and Interface

As illustrated in Fig. 4, *TacMagPie* operates as a plug-and-play backend to a physics simulator. The physics simulator is responsible for scene management, collision detection, and dynamics solving, while *TacMagPie* focuses on elastomer deformation and magnetic field computation. By adjusting the hyperelastic parameters and Hall sensor positions in the configuration file, the framework can be adapted to different sensor designs. Regarding simulator compatibility, we currently provide an interface for MuJoCo; however, any physics simulator that supports the interaction pipeline depicted in Fig. 4 can be integrated with our simulator.

IV. EXPERIMENTAL VALIDATION SETUP

This section describes the dataset used to evaluate *TacMagPie*, covering both the real-world data collection apparatus and the corresponding simulation configuration.

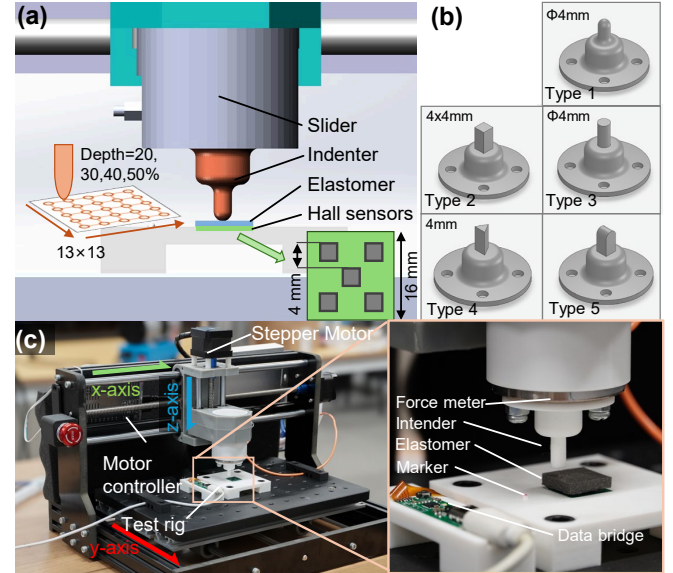


Fig. 5. Experimental setup. (a) A slider-driven indenter presses into the elastomer at specified locations and depths, five Hall sensors measure magnetic flux. (b) Five indenter geometries evaluated in this study. (c) Real-world setup for sMagTac, including an automated calibration platform with force-torque sensor and motorized stage.

A. Benchmark Dataset

To validate *TacMagPie* against physical measurements, a custom sMagTac was fabricated. The sensor consists of a 16×16 mm silicone pad (Ecoflex 00-30, 2 mm thick) doped with N35 NdFeB microparticles (100 mesh) and magnetized to saturation along the z -axis. Five triaxial Hall-effect sensors (MLX90393, Melexis, Belgium) are mounted on a PCB beneath the elastomer, with four placed at the corners of an 8 mm square and one at the center, sampling magnetic flux density $\mathbf{B}_{\text{real}}^M$ at 50 Hz.

A benchmark dataset was then constructed using a 3-axis motorized platform equipped with a force-torque sensor

(γ75D, Decent, China) and five interchangeable indenters of distinct geometries, as shown in Fig. 5(b)(c). For each indenter, the probe scanned a 12×12 mm contact region at 1 mm spatial resolution, with indentation depths ranging from 20% to 50% of the elastomer thickness. Each measurement was repeated 20 times and then averaged, yielding a measured response tensor $\mathcal{D}_{\text{real}} \in \mathbb{R}^{13 \times 13 \times N_{\text{depth}} \times 5 \times 3}$ per indenter.

B. Simulation Configuration

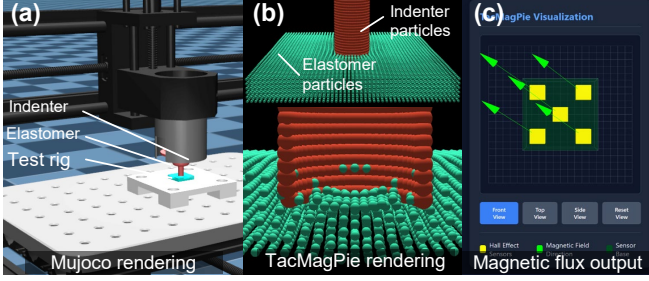


Fig. 6. Simulator interface consists of three components: (a) the MuJoCo rendering window, which displays the full scene geometry and object interactions; (b) the *TacMagPie* particle viewer, which visualizes the discretized elastomer and interacting object as material point clouds; and (c) the virtual Hall sensor output panel, which displays the real-time magnetic field readings from each simulated sensor element.

Fig. 6(a)-(c) show the MuJoCo interface connected to *TacMagPie*, the particle rendering backend, and the magnetic signal interface, respectively. The MuJoCo scene comprises a test platform, an elastomer body, and a spherical indenter attached to a prismatic joint driven by a position controller for uniform-velocity indentation. Since elastomer reaction forces are handled entirely by *TacMagPie*, the elastomer is initialized in MuJoCo as a passive rigid body with mass but no collision geometry.

Within *TacMagPie*, both the elastomer and the interacting indenter are discretized within a shared particle simulation domain of $20 \times 20 \times 20$ mm, resolved on a $32 \times 32 \times 32$ grid with 8 particles per cell. Indenter particles are generated from a point cloud via Open3D, while the elastomer is represented by a fixed $51 \times 51 \times 6 = 15,606$ -particle grid. Grid and particle densities are configurable; finer resolutions improve accuracy at the cost of increased computation time.

Elastomer material properties are adopted from [25], with Yeoh parameters $C_1 = 1.00 \times 10^{-1}$, $C_2 = 6.42 \times 10^{-2}$, $C_3 = 8.08 \times 10^{-5}$ and density $\rho = 1.07 \times 10^3$ kg/m³. To model the viscoelastic dissipation of silicone, a damping factor $\alpha = 0.99$ is applied to all particles at each time step.

V. EXPERIMENTAL RESULTS

In this section, we evaluate the performance of *TacMagPie* to answer three primary research questions:

Q1: How accurately does the simulator compute magnetic flux compared to real-world measurements? (see Sec. V-A)

Q2: Is the simulator fast enough to support real-time applications? (see Sec. V-B)

Q3: Can models trained in simulation generalize to the physical world for perception tasks? (see Sec. V-C)

A. Fidelity of Magnetic Flux Reconstruction

To quantitatively assess the physical consistency of *TacMagPie*, we compare three simulation approaches against the benchmark dataset: a mass-spring model (via MuJoCo’s FlexComp component), FEM, and our method. All baselines use the same sensor geometry, Hall sensor locations, indentation trajectories, and evaluation protocol as *TacMagPie*. Their parameters were selected to obtain stable responses under practical runtime and memory budgets, rather than through exhaustive hyperparameter or mesh-convergence search. Unlike vision-based tactile sensors that output RGB images, sMagTac produce magnetic flux vectors. To enable a unified evaluation, the outputs of all three methods are organized in the same format as the real dataset described in Sec. IV-A, yielding tensors of shape $\mathbb{R}^{13 \times 13 \times N_{\text{depth}} \times 5 \times 3}$. This formulation allows standard image-quality metrics to be applied for quantitative analysis.

As shown in Table II, three metrics are used to evaluate the discrepancy between simulated and real measurements. Root Mean Square Error (RMSE) measures absolute error; given that the real sensor operates over a response range of approximately $[-2000, 2000]$ μT , this metric provides a physically meaningful error scale. Peak Signal-to-Noise Ratio (PSNR) assesses global response consistency, while Structural Similarity Index Measure (SSIM) evaluates spatial structural similarity, reflecting the simulation’s ability to reproduce the fine-grained distribution of the magnetic field.

The experimental results demonstrate that *TacMagPie* achieves substantially higher physical consistency in reconstructing magnetic flux density compared to the baseline methods.

The mass-spring model, due to its overly simplified linear elasticity assumption, fails to capture the nonlinear response of silicone. Furthermore, its numerical instability at higher particle counts frequently causes the elastomer simulation to diverge, resulting in RMSE exceeding 1000 μT , indicating that this baseline is insufficient to reproduce the spatial structure of the magnetic flux response.

FEM provides a physically meaningful reference for non-linear elastic deformation, but in our practical configuration its accuracy is limited by the trade-off between resolution, runtime, and memory cost; this comparison does not imply that FEM is intrinsically less accurate than *TacMagPie*. The resulting RMSE falls in the range of 200–300 μT , PSNR is approximately 19 dB, and SSIM lies between 0.25 and 0.45, indicating notable structural distortion.

In contrast, *TacMagPie* achieves an RMSE of 40–70 μT , PSNR exceeding 25 dB, and SSIM of 0.56–0.68, demonstrating that the simulation accurately captures both the spatial distribution and response trends of the magnetic field while maintaining real-time performance.

Fig. 7 provides a qualitative visual comparison of the three methods against real-world measurements. While minor high-frequency noise and localized discrepancies are present, the overall spatial patterns and response trends are in close agreement, validating the physical fidelity of *TacMagPie* in reproducing the behavior of magnetic tactile sensors.

TABLE II
QUANTITATIVE EVALUATION OF SIM-TO-REAL FIDELITY ACROSS DIFFERENT INDENTERS AND THICKNESSES

Method	Depth (%)	Type 1			Type 2			Type 3			Type 4			Type 5		
		RMSE ↓	PSNR ↑	SSIM ↑	RMSE ↓	PSNR ↑	SSIM ↑	RMSE ↓	PSNR ↑	SSIM ↑	RMSE ↓	PSNR ↑	SSIM ↑	RMSE ↓	PSNR ↑	SSIM ↑
Mass spring	20	>1e3	13.57	0.01	>1e3	13.46	0.03	>1e3	13.36	-0.14	>1e3	13.33	-0.09	>1e3	13.20	-0.09
	30	>1e3	13.61	-0.05	>1e3	13.80	-0.14	>1e3	13.87	-0.19	>1e3	13.33	-0.11	>1e3	13.33	-0.20
	40	>1e3	14.02	-0.01	>1e3	13.75	-0.01	>1e3	14.20	-0.18	>1e3	13.45	-0.14	>1e3	13.35	-0.17
	50	>1e3	14.17	0.00	>1e3	14.06	0.02	>1e3	13.87	0.26	>1e3	13.43	0.16	>1e3	13.29	0.25
FEM	20	273.75	19.14	0.35	253.08	19.30	0.34	214.89	19.19	0.30	237.48	19.07	0.29	236.58	19.28	0.25
	30	251.73	19.28	0.36	248.02	19.44	0.33	261.75	19.87	0.34	231.49	20.22	0.29	212.56	20.04	0.33
	40	261.85	19.44	0.38	255.09	19.00	0.39	263.77	19.37	0.36	241.28	19.29	0.33	284.19	19.03	0.37
	50	244.02	19.45	0.38	249.73	19.48	0.42	274.46	19.09	0.43	233.63	19.11	0.44	228.21	19.56	0.40
TacMagPie (ours)	20	61.96	24.72	0.59	69.75	25.31	0.60	60.56	24.85	0.60	51.52	25.06	0.62	56.96	25.39	0.61
	30	63.22	25.44	0.62	65.60	26.01	0.64	58.27	25.95	0.63	55.06	25.54	0.65	57.02	25.95	0.66
	40	40.80	25.24	0.64	41.81	25.24	0.68	65.23	25.07	0.64	63.12	25.90	0.66	56.68	25.55	0.66
	50	70.37	25.30	0.65	49.43	25.66	0.67	60.73	26.12	0.63	44.15	25.76	0.65	68.38	26.21	0.68

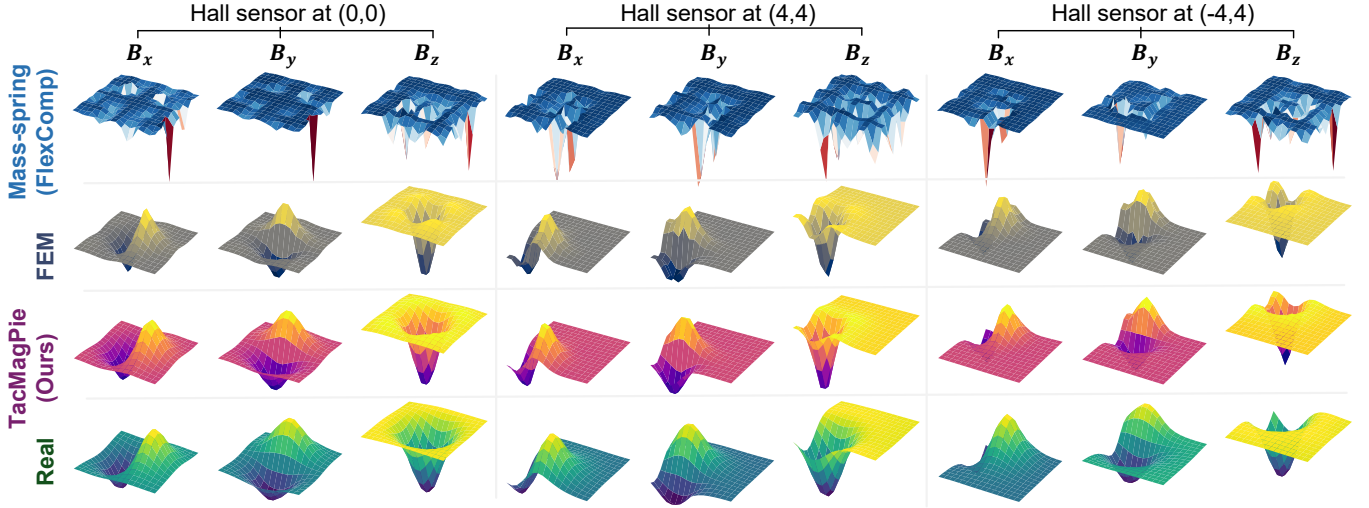


Fig. 7. Visual comparison of magnetic flux maps reconstructed by the three simulation methods against real-world measurements. Results are shown for the Type-1 indenter at 50% indentation depth, with columns corresponding to Hall sensor responses at horizontal positions (0, 0), (4, 4), and (−4, 4) mm, from left to right. Note that these dense maps are not spatial magnetic-field images measured by a dense Hall array; each pixel corresponds to one controlled indentation location in the 13×13 scan grid.

B. Computational Efficiency and Real-Time Performance

We further evaluate the computational efficiency of *TacMagPie* in terms of average step time and memory footprint, as reported in Table III.

TABLE III
COMPUTATION TIME AND MEMORY USAGE PER SIMULATION STEP

Method	Step Time (ms)	Memory (MB)
TacMagPie (CPU)	2.45	155
TacMagPie (GPU)	0.32	246
Mass-Spring	24	1700
FEM	2150	3900

FEM incurs a per-step cost of 2150 ms and requires 3900 MB of memory, making it difficult to use as a repeated simulation backend for real-time interaction or large-scale data generation. The mass-spring model, despite its structural simplicity, consumes 1700 MB and averages 24 ms per step due to MuJoCo’s internal friction dynamics, also falling short of real-time requirements.

TacMagPie achieves a per-step time of 2.45 ms on CPU (Intel Core I9-13900K) with only 155 MB of memory, already representing a $\sim 880\times$ speedup over FEM. With GPU acceleration (NVIDIA RTX 4090D), the step time is further reduced to 0.32 ms, nearly 6700 $\times$ faster than FEM, while memory usage remains modest at 246 MB. This level of efficiency makes *TacMagPie* well-suited for large-scale synthetic data generation and synthetic tactile data generation pipelines.

C. Synthetic-to-Real Contact Localization

To assess the simulator’s utility for downstream learning, we conduct a “Contact Position Super-Resolution” task, which requires the model to infer continuous contact coordinates (x, y) from the sparse readings of 5 discrete Hall sensors. We formulate this as a regression problem, where a lightweight MLP with 3 hidden layers (of sizes 64, 128, and 64) maps the magnetic flux deviations $\Delta \mathbf{B} \in \mathbb{R}^{15}$ to the 2D contact position. Three distinct training protocols are evaluated:

- **Real2Real (Real-data baseline):** The network is

trained on 80% of the real-world dataset $\mathcal{D}_{\text{real}}$ and tested on the remaining 20%. This serves as a real-data-only reference baseline.

- **Sim&Real2Real (Hybrid):** The network is trained on the synthetic dataset $\mathcal{D}_{\text{sim}}$ generated by *TacMagPie* and 80% of $\mathcal{D}_{\text{real}}$, and evaluated on the held-out 20%. This protocol assesses whether simulation data can augment real-world learning.
- **Sim2Real (Zero-Shot):** The network is trained *exclusively* on the synthetic dataset $\mathcal{D}_{\text{sim}}$ and then directly evaluated on the real-world test set.

TABLE IV
SIM-TO-REAL CONTACT LOCALIZATION PERFORMANCE

Training Protocol	Indenter Type	Localization Error (mm)
Real2Real	Type 1	0.63 ± 0.23
	Hybrid (1-5)	0.49 ± 0.15
Sim&Real2Real	Type 1	0.58 ± 0.13
	Hybrid (1-5)	0.47 ± 0.15
Sim2Real	Sphere (Type 1)	2.56 ± 0.73
	Hybrid (1-5)	2.60 ± 0.83

After averaging 20 repeated measurements, the real dataset contains 676 samples per indenter and 3380 samples in total across five indenters. The results are summarized in Table IV. The *Real2Real* baseline achieves sub-millimeter precision (≈ 0.6 mm), confirming the intrinsic high sensitivity of the hardware. Crucially, the *Sim2Real* models, despite having never been exposed to the physical sensor, achieve a localization accuracy of ≈ 2.6 mm. Meanwhile, the “Sim&Real2Real” protocol marginally surpassed the “Real2Real” baseline.

While a performance gap exists due to the inevitable Reality Gap (fabrication tolerances, imperfect sensor alignment, and demagnetization effects not fully captured in the current model), the network successfully learns the correct underlying inverse physics mapping. This result confirms that *TacMagPie* provides sufficiently high-quality synthetic data to support Zero-Shot transfer for spatial perception tasks, validating its effectiveness as a training ground for learning-based algorithms.

VI. CONCLUSION AND FUTURE WORKS

We presented *TacMagPie*, a real-time and physically consistent simulator for soft magnetic tactile sensors. By coupling MLS-MPM-based hyperelastic deformation modeling with the proposed Differential Magnetic Microparticle Method (DMM), *TacMagPie* efficiently reconstructs magnetic tactile responses from elastomer deformation. Experimental results on controlled indentation benchmarks show strong agreement with real-world Hall-sensor measurements, achieving a PSNR of 26.21 dB and an SSIM of 0.68. Meanwhile, the GPU implementation runs in under 1 ms per simulation step, up to $6700\times$ faster than FEM. These results demonstrate the potential of *TacMagPie* as a practical simulation backend for synthetic tactile data generation and learning-based magnetic tactile perception.

The current validation focuses on controlled quasi-static indentation, which allows precise evaluation of magnetic tactile response reconstruction but does not fully cover dynamic, shear-dominant, or task-level robotic manipulation scenarios. In addition, the present noise model mainly accounts for additive electronic noise, while more complex effects such as hysteresis and long-term drift remain to be incorporated. Future work will extend *TacMagPie* to richer contact conditions, more hardware-specific sensor models, and real robotic manipulation tasks.

To facilitate further research, the full codebase is open-sourced at <https://github.com/ultralyj/TacMagPie>.

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